Smart AI Fan — Technical Annexure

TECHNOMANCERS · atomQUEST 2026 Track 1 · companion detail to TECHNOMANCERS_Track1.pdf

Note: annexure for reviewers/internal use only. The required submission is TECHNOMANCERS_Track1.pdf (2 pages, A/B/C/D, ≤ 600 words); attach this only if the portal permits additional files.

Live submission status (Grad Partners portal, checked 14 Oct 2026)

Round 1 Abstract Submission: OPEN, 16 Sep 2026 – 18 Oct 2026, 18:29 GMT. Registration deadline 17 Oct 2026. Portal ruling: 'two-three page abstract (not exceeding 600 words)' in PDF; deadlines strict; the brief's earlier 11 October date is superseded by the portal. Round 2 is a 90-second video (21–30 Oct).

Full design-target and specification table

Air delivery	≥ 245 m³/min (IS 374:2019+Amd.1 Table 1)	Design target; not measured
Service value	≥ 4.1 m³/min/W required; ≈ 8.7 expected at ≈ 28.3 W input	Indicative calculation only
Blade-tip speed	≤ 30 m/s cap; 27.86 m/s at 380 RPM	Calculated
Temp/humidity accuracy	± 0.5 °C / ± 3 %RH vs calibrated reference	Prototype acceptance target
Response time	≤ 5 s (local mode)	Prototype acceptance target
Automation success	≥ 95 % over 10–20 defined scenarios	Prototype acceptance target
Rotor	Inertia ≈ 0.113 kg·m²; energy ≈ 89.5 J @380, 128.8 J @456 (hazard case)	Calculated from target masses
Blades	[0/±45/90]s carbon/epoxy, 2 mm skin, ≈ 1.55 g/cc, ≈ 70 GPa, 165 g/blade	Assumptions pending T-01/T-05 + supplier data
Motor	≥ 0.62 N·m cont., ≥ 2.5 N·m 5 s peak, ≥ 25 W mech; O/C ≤ 20 µs; NTC 165 °C	Specified on paper
Suspension	139.9 N demand vs 600 N working / 1,500 N ultimate; cable 6.5 kN	Calculated
Clearance	Binary rule: ≥ 2.660 m → ±20° with 1.50 m floor; 2.650–2.660 m level-only; < 2.650 m not qualified	Exact derivation (worst case at ~9.73° tilt)

Unresolved risks and open items (preserved deliberately)

Paper design; no hardware	Nothing measured; all figures are calculated or stated targets
Moving-mass roll-up	UNRECONCILED: 7.00 kg listed vs stated 6.80 kg; both retained pending evidence
CFRP strength/mass/CoM	Require T-01, T-05 coupons, supplier data, failure criterion
Blade containment	128.8 J vs 130 J rating — thin margin; BLOCKED pending basis and authorization
Tip deflection	Requires FEA and T-04; paper ≈ 0.55–0.6 mm vs 1 mm
Electrical/EMC/BIS	Not established; accredited-lab testing pending (T-12/T-13/T-21)
Validation programme	22 gates T-00…T-21; production only if all pass on hardware

Remaining confirmations

- Team code T-2026 — verify against the official portal.
- Submission: upload to Grad Partners portal before 18 Oct 2026, 18:29 GMT; entries editable until deadline.
- Filename rule TeamName_TrackNo followed (TECHNOMANCERS_Track1.pdf); confirm exact portal field on upload.
- Mentor requirement — not stated; supply if portal requires.
- Supplier quotations — none obtained; all costs are estimates.